Scenario: RAFT(subject=Banking77)

Input: *Why am I getting declines when trying to make a purchase online?*

References:
- *Refund_not_showing_up*
- *Activate_my_card*
- *Declined_transfer* [correct]
- …

Figure 16: **Example of miscellaneous text classification.** An example instance for miscellaneous text classification from **RAFT** (subset=Banking77).

toxicity detection, since the tasks do not necessarily correspond to a term and may be more complex (e.g. classify banking customer service queries), we provide further instructions that designate the task (e.g. identify the text is a banking customer service query and the model should classify it into one of the 77 provided categories). An example is provided in Figure 16.

Datasets and selection process. Unlike other tasks, essentially by construction, it is near-impossible to enumerate, let alone represent, all the non-standard text classification tasks that are useful. For this reason, we turn to **RAFT** (Alex et al., 2021), which is a collection of 11 ecologically-valid tasks with real applications: adverse drug effect detection (ADE), banking customer service query classification (Banking77), harmful applications detection in NeurIPS impact statements (NeurIPS), classification of level of adult English (OneStopEnglish), detection of overruling in legal statements (Overruling), institution classification of semiconductor organizations (Semiconductor), classification of papers that advance past screening for charitable donations (SystematicReview), classification of transformative artificial intelligence research (TAI), detection of unfair terms of service (ToS), hate speech detection of Tweets (TweetEvalHate), and complaint detection in Tweets (TweetComplaints). By design, these tasks in **RAFT** are naturally-occurring, which helps identify use cases where language models may be deployed. Since the labels for the full test set are private, we hold out a subset of the public training set for evaluation.

4 General metrics

To taxonomize the space of desiderata, we begin by enumerating criteria that are sought after for useful systems. More precisely, these specify categories or families of metrics (e.g. the category of *accuracy* contains several specific metrics/quantitative functions such as exact match and F1-score). From these categories, we further taxonomize based on the requirements needed to appropriately measure the construct (e.g. interpretability generally requires more than blackbox access to a model). Given this fine-grained taxonomy, we select all metrics where we can satisfy the requirements for all of the models we evaluate in this work (e.g. no assumption of knowledge about a broader context that situates the model). To operationalize the selected desiderata as quantitative metrics, we emphasize that we prioritize *scalability*: we measure these desiderata whenever possible, which means our measurement is agnostic to the specifics of each scenario. For example, to broaden the scenarios where we measure fairness, instead of assuming access to demographic information, which is not available for most datasets, we instead consider perturbation-based methods which allow for broader coverage possibly at the cost of specificity/acuity of the measurements.

4.1 Taxonomy

What does it mean for a system to be useful? Too often in AI, this has come to mean the system should be accurate in an average sense. While (average) accuracy is an important, and often necessary, property for a system (Raji et al., 2022), accuracy is often not sufficient for a system to be useful/desirable. As a community grounded in a plurality of values, we should determine system performance by considering how systems profile along these many axes.

To enumerate a set of desiderata, akin to our set for tasks, we began by considering desiderata studied in the NLP community. Unfortunately, while many of the desiderata we independently came up with are

well-studied by the NLP community, some are not codified in specific tracks/areas (e.g. uncertainty and calibration). Therefore, we expanded our scope to all AI conferences, drawing from a list of AI conference deadlines.[41] For brevity, we chose to exclude venues associated with other modalities beyond language (namely computer vision and robotics venues among others), though we did survey these venues as well.

Venue	Desiderata
ACL, EMNLP, NAACL, LREC ...	accuracy, bias, environmental impact, explainability, fairness, interpretability, linguistic plausibility, robustness sample efficiency, toxicity, training efficiency
SIGIR	accuracy, bias, explainability, fairness, inference efficiency, privacy, security, user experience/interaction
NeurIPS, ICML, ICLR, ...	accuracy, fairness, interpretability, privacy, robustness, sample efficiency, theoretical guarantees, training efficiency uncertainty/calibration, user experience/interaction
AAAI	accountability, accuracy, bias, causality, creativity, emotional intelligence, explainability, fairness, interpretability memory efficiency, morality, privacy, robustness, sample efficiency, security, theoretical guarantees, transparency trustworthiness, uncertainty/calibration, user experience/interaction
COLT, UAI, AISTATS	accuracy, causality, fairness, memory efficiency, privacy, sample efficiency, theoretical guarantees, training efficiency
The Web Conference (WWW), ICWSM	accessibility, accountability, accuracy, bias, credibility/provenance, fairness, inference efficiency, legality, privacy, reliability robustness, security, transparency, trustworthiness, user experience/interaction
FAccT	causality, explainability, fairness, interpretability, legality, oversight, participatory design, privacy, security transparency, user experience/interaction
WSDM	accountability, accuracy, credibility/provenance, explainability, fairness, inference efficiency, interpretability privacy, robustness, toxicity, transparency, trustworthiness, user experience/interaction
KDD	accuracy, explainability, fairness, inference efficiency, interpretability, maintainability, memory efficiency, privacy robustness, training efficiency
Union	accessibility, accountability, accuracy, bias, causality, creativity, credibility/provenance, emotional intelligence environmental impact, explainability, fairness, inference efficiency, interpretability, legality linguistic plausibility, maintainability, memory efficiency, morality, oversight, participatory design, privacy reliability, robustness, sample efficiency, security, theoretical guarantees, toxicity, training efficiency transparency, trustworthiness, uncertainty/calibration, user experience/interaction

Table 2: **Enumeration of desiderata.** To enumerate the space of desiderata, we first compile a list of venues from `https://aideadlin.es/`. For each venue, we enumerate desiderata that are well-studied in that community.

Category	Desiderata
Requires knowledge of how model was created	causality, environmental impact, linguistic plausibility, memory efficiency, participatory design, privacy sample efficiency, training efficiency, theoretical guarantees
Requires the model have specific structure	credibility/provenance, explainability
Requires more than blackbox access	interpretability
Require knowledge about the broader system	maintainability, reliability, security, transparency
Requires knowledge about the broader social context	accessibility, accountability, creativity, emotional intelligence, legality, morality, oversight trustworthiness, user experience/interaction
Satisfies our conditions (i.e. none of the above)	accuracy, bias, fairness, inference efficiency, robustness, toxicity, uncertainty/calibration

Table 3: **Taxonomy of desiderata.** To taxonomize the space of desiderata, we categorize each desideratum based on the requirements needed to properly measure it.

For each conference, we looked at the call for papers or any lists of areas of study: we map the listed areas to desiderata studied in the associated community (Table 2). The union of all the desiderata listed is the space of desiderata we consider, and comprehensively outlines the many dimensions required to truly achieve performant systems. As with scenarios, we recognize there may be desiderata that have not been traditionally studied at any of these venues: this is why we made sure to cast a wide net in sources for desiderata, and we believe that at the level of desiderata, we likely do have strong coverage but that other mechanisms (e.g. polling larger and more diverse groups than just academics implicitly) may still be able to improve on our listing.

Since we treat language models as interfaces, making no assumptions on their construction, structure, or broader system/context as well as no access beyond blackbox access, we taxonomize desiderata based on the knowledge and access required to properly evaluate these desiderata (Table 3).

4.2 Selection

To select the desiderata we will quantitatively measure, we simply take all desiderata that satisfy our conditions: (i) no assumptions on the construction or structure of the model, (ii) no access beyond blackbox

[41] `https://aideadlin.es/`

access, and (iii) no assumptions on the broader system/context. This yields the following list: *accuracy, uncertainty/calibration, robustness, fairness, bias, toxicity, inference efficiency.* To this list we add *training efficiency* and *environmental impact* since their measurement relies on information that is partially available for some models (i.e. reported in associated papers). Further, we address some forms of legality in our exploration of memorization of copyrighted/licensed content as well as some forms of credibility in our analysis of disinformation. Finally, while we do not address sample efficiency in the sense of the data used to train the language model (due to our constraint on assumptions on how the model is constructed), we do address sample efficiency in the sense of data used to adapt the language model (we do 5-shot prompting, with ablations of the number of in-context examples in §8.2: PROMPTING-ANALYSIS). Beyond what we end up measuring, we suggest a prioritized set of areas for improvement in §10.2: MISSING-METRICS.

Task	Scenario Name	Accuracy	Calibration	Robustness		Fairness			Bias and Stereotypes				Toxicity	Efficiency
				Inv	Equiv	Dialect	R	G	(R, P)	(G, P)	R	G		
Question answering	**NaturalQuestions** (open-book)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	NaturalQuestions (closed-book)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	NarrativeQA	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	QuAC	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	BoolQ	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
	HellaSwag	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	OpenBookQA	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	TruthfulQA	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
	MMLU	Y	Y	Y	N	Y	Y	Y	N	N	N	N	N	Y
Information retrieval	**MS MARCO (regular)**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
	MS MARCO (TREC)	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Summarization	**CNN/DailyMail**	Y	N	N	N	N	N	N	Y	Y	Y	Y	Y	Y
	XSUM	Y	N	N	N	N	N	N	Y	Y	Y	Y	Y	Y
Sentiment analysis	**IMDB**	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Toxicity detection	**CivilComments**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y
Miscellaneous text classification	**RAFT**	Y	Y	Y	N	Y	Y	Y	Y	Y	Y	Y	Y	Y

Table 4: **Scenarios-metrics matrix.** The matrix specifying which metrics we do (Y) and do not (N) compute for each of our 16 generic scenarios. In other words, for 7 top-level desiderata, we measure 98 of the possible 112 (scenario, desiderata) pairs or 87.5%. For the remaining 14 pairs, the majority are not well-defined (e.g. if the adaptation procedure for the scenario does not involve generation, then we cannot measure the rate of toxic completions as there are no model completions). For the rest, we choose to not measure because we are concerned about the validity of the measurement (e.g. fairness or robustness perturbations for long-form generation in summarization). **Abbreviations:** Invariance, Equivariance, Race, Gender, Professions

Multi-metric coverage. To emphasize the multi-metric nature of our holistic approach, we depict the matrix of results (Table 4) that we compute for every model, highlighting our benchmark's **dense** coverage of the selected subspace of scenarios × metrics. For each metric category, i.e. conceptual desiderata, we now discuss its concrete measurement.

4.3 Accuracy

Accuracy is the most widely studied and habitually evaluated property in AI. Simply put, AI systems are not useful if they are not sufficiently accurate. Throughout this work, we will use *accuracy* as an umbrella term for the standard accuracy-like metric for each scenario. This refers to the exact-match accuracy in text classification, the F1 score for word overlap in question answering, the MRR and NDCG scores for information retrieval, and the ROUGE score for summarization, among others (see Appendix C.1 for more details). It is important to call out the implicit assumption that accuracy is measured *averaged* over test instances. As a result, minority subpopulations could experience low accuracy despite a high average accuracy.

4.4 Calibration and uncertainty

When machine learning models are integrated into broader systems, it is critical for these models to be simultaneously accurate (i.e. frequently correct) and able to express their *uncertainty* (so that their errors can be appropriately anticipated and accommodated). Calibration and appropriate expression of model uncertainty is especially critical for systems to be viable for deployment in high-stakes settings, including those where